



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2016
Higher 2

BIOLOGY

9648/01

Paper 1 Multiple Choice

Tuesday, 20 September 2016

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number, index number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C, D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

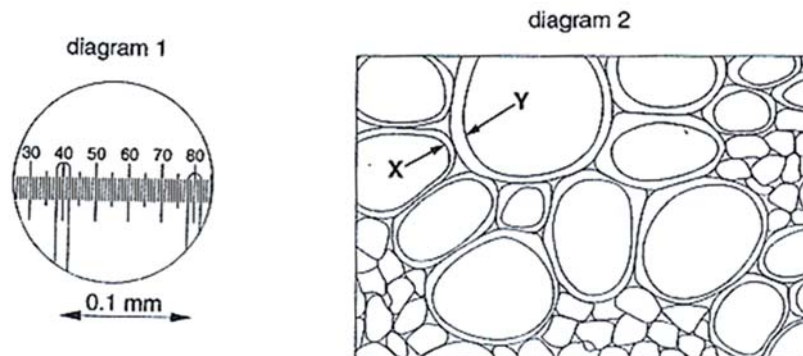
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough work should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

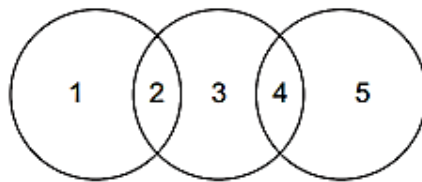
- 1 Diagram 1 shows a stage micrometer, on which the graduations are 0.1mm apart, and an eyepiece graticule, as seen through a light microscope.

Diagram 2 shows a section of plant tissue viewed at the same magnification.



What is the **approximate total thickness** of the cell walls between X and Y?

- A $10\ \mu\text{m}$ **B $20\ \mu\text{m}$** C $100\ \mu\text{m}$ D $200\ \mu\text{m}$
- 2 The diagram shows some **similarities** between **chloroplasts**, **mitochondria** and **typical prokaryotes**.



Which of the following is **correct**?

	1	2	3	4	5
A	chloroplasts	circular DNA	mitochondria	80s ribosomes	prokaryotes
B	chloroplasts	80s ribosomes	mitochondria	circular DNA	prokaryotes
C	prokaryotes	circular DNA	mitochondria	circular DNA	chloroplasts
D	prokaryotes	70s ribosomes	chloroplasts	80s ribosomes	mitochondria

3 The table below shows the results of experiments on the rates of absorption of some monosaccharide sugars by pieces of living small intestine, and by similar pieces of intestine poisoned with cyanide which affects the electron transport chain.

sugar	rate of absorption/ arbitrary units	
	by living intestine	by poisoned intestine
glucose	1.00	0.33
xylose	0.30	0.31
fructose	0.40	0.37
galactose	1.10	0.53
arabinose	0.29	0.29

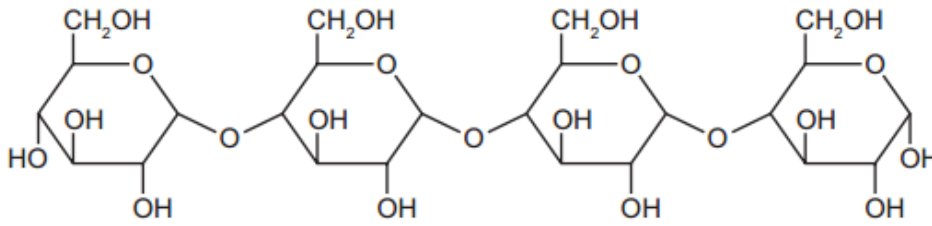
Which sugars are being actively absorbed?

- A glucose and fructose
- B glucose and galactose
- C glucose and xylose
- D xylose and arabinose

4 A marine mammal, such as a seal or porpoise, stores large amounts of subcutaneous fat as 'blubber'. Which one of the following is an incorrect statement about the adaptive value of this fat?

- A It has a low energy content for its mass in comparison to other food reserves such as glycogen.
- B It has an insulating function, fat having a low heat conductivity.
- C It acts as an energy store, being utilised when the animal is short of food.
- D It is insoluble in water therefore it does not affect the osmotic balance of the cells.

- 5 The molecule shown is a **polymer** of reducing sugars.

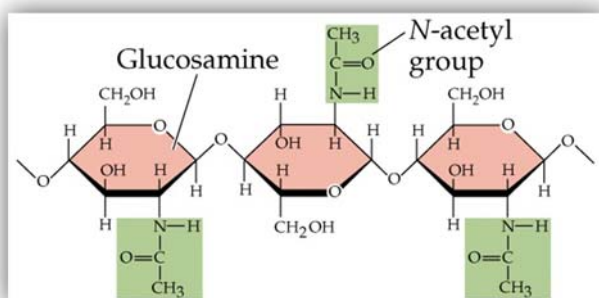


Which **procedures** could be carried out in order to **test** for the **presence** of **reducing sugars** in this molecule?

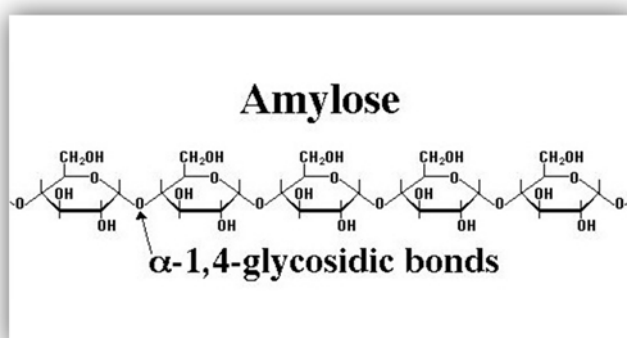
- 1 Add hydrolytic enzyme and then heat with Benedict's reagent.
 - 2 Dissolve in water, neutralize and then heat with Benedict's reagent.
 - 3 Boil with hydrochloric acid, neutralize and then heat with Benedict's reagent.
- A** 1, 2 and 3
B 1 and 2 only
C 1 and 3 only
D 2 and 3 only

6 An **unusual** enzyme has been found in a tropical grass.

- It catalyses the **hydrolysis** of the fungal **polysaccharide**, **chitin**, **into** **amino sugars**.



- It also **inhibits** the **activity** of an **enzyme** in locust guts which **catalyses** the **digestion** of **amylose**.

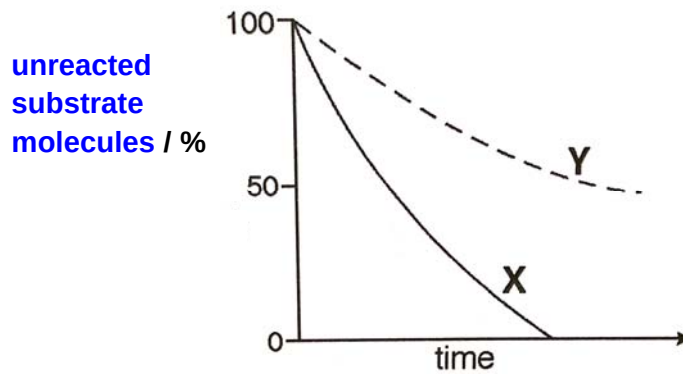


Which of the following **describes** the **actions** of this **unusual enzyme**?

	reaction catalysed	reaction inhibited
A	hydrolysis of glycosidic bonds	condensation of glycosidic bonds
B	hydrolysis of glycosidic bonds	hydrolysis of glycosidic bonds
C	hydrolysis of peptide bonds	condensation of glycosidic bonds
D	hydrolysis of peptide bonds	hydrolysis of glycosidic bonds

- 7 Curve X represents the course of an enzyme-catalysed reaction under optimum conditions.

Curve Y shows the action of the same enzyme on the same substrate but with one alteration to the reaction conditions.



Which factor, operating to a constant extent throughout the experiment, could give the results shown by curve Y?

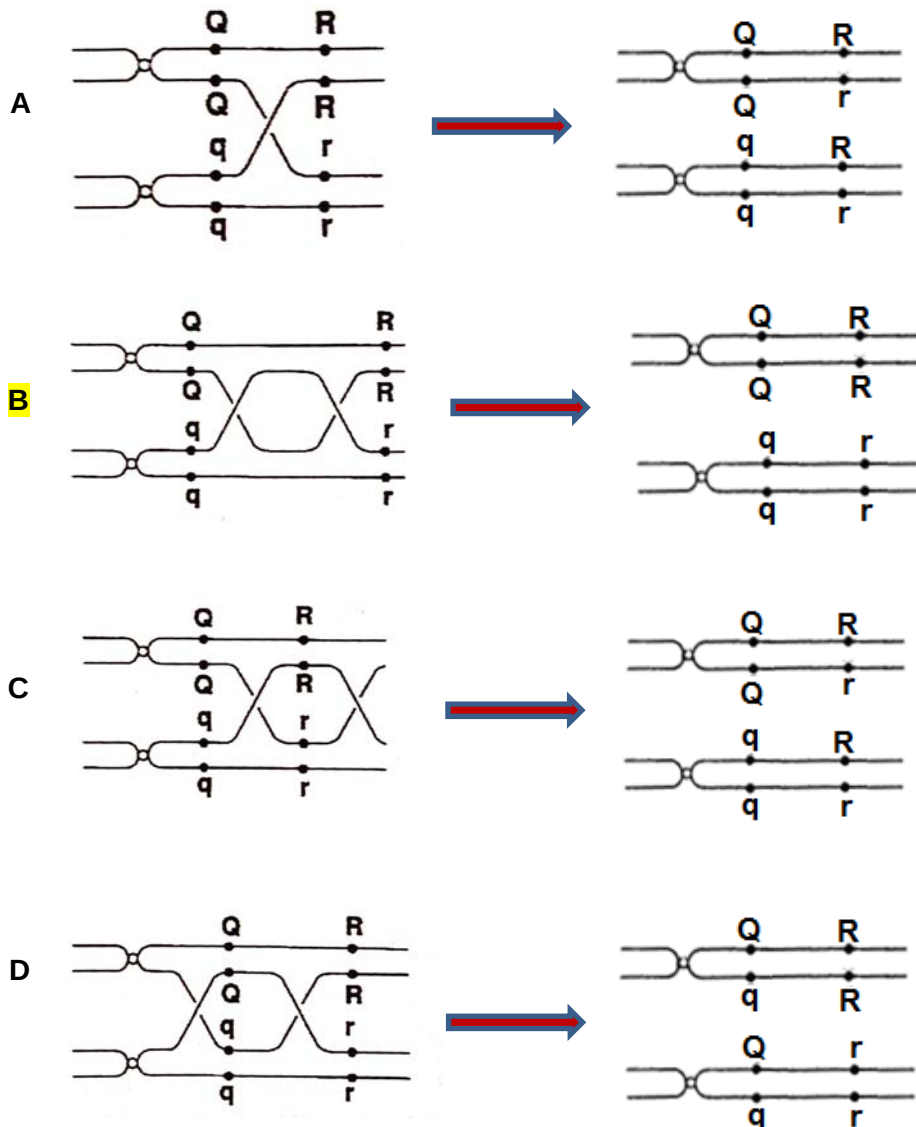
- A increased substrate concentration
 - B inhibition by the end product**
 - C less concentrated reaction mixture
 - D lower temperature
- 8 The diagram shows some chromosomes at late prophase of mitosis.



How many chromosomes would be present in one nucleus at telophase II of meiosis?

- A 6**
- B 12
- C 18
- D 24

- 9 Which diagram represents the situation which would result in **segregation** of the alleles **Q** from **q**, and **R** from **r** in the **first meiotic division**?



- 10 Cells from a bacterial clone were grown for many generations on a medium in which the nitrogen compounds contained only the isotope nitrogen 15 (^{15}N). **Adenine** comprised **36%** of the nitrogen bases present. A sample of these bacteria was transferred to a medium in which the only nitrogen source was ^{14}N and was provided with conditions suitable for asexual reproduction.

What was the **percentage** of **guanine** in the **DNA**?

- A** 14%
- B** 18%
- C** 28%
- D** 36%

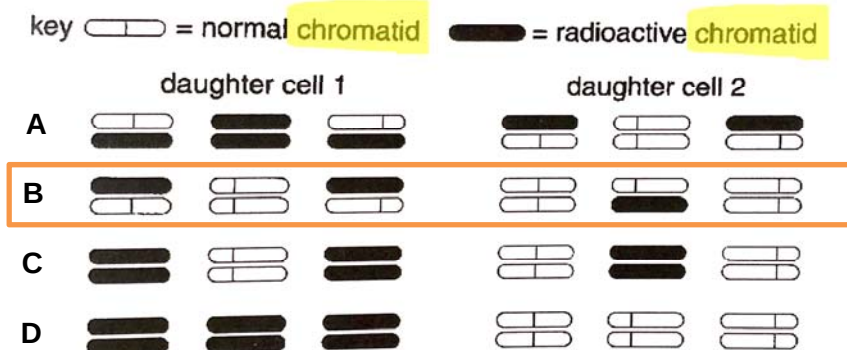
- 11 If there were 34 amino acids and DNA only contained two types of nitrogenous bases, what would be the minimum number of bases per codon that could code for proteins?

A 3 B 4 C 6 D 8

- 12 The following experiment was carried out.

- 1 Haploid cells, containing three chromosomes each, were grown in a medium containing radioactive thymine, so that all the DNA was labelled.
- 2 Cells in early interphase were then transferred to a medium where the available thymine was not radioactive.
- 3 A single cell was immediately isolated and allowed to divide once. When the two daughter cells reached the next metaphase, they were fixed and their three chromosomes were inspected for radioactivity.

Which diagram represents the distribution of radioactivity at metaphase in the two daughter cells?



- 13 An antibiotic, edeine, was isolated. It inhibits protein synthesis but has no effect on either DNA synthesis or RNA synthesis. When added to a translation mixture containing fully intact organelles, edeine stops haemoglobin translation after 10 seconds.

Analysis of the edeine-inhibited mixture by centrifugation showed that no ribosomes remained by the time protein synthesis had stopped. Instead, all the mRNA accumulated, together with small ribosomal subunit and initiator tRNA.

Which step in protein synthesis does edeine inhibit?

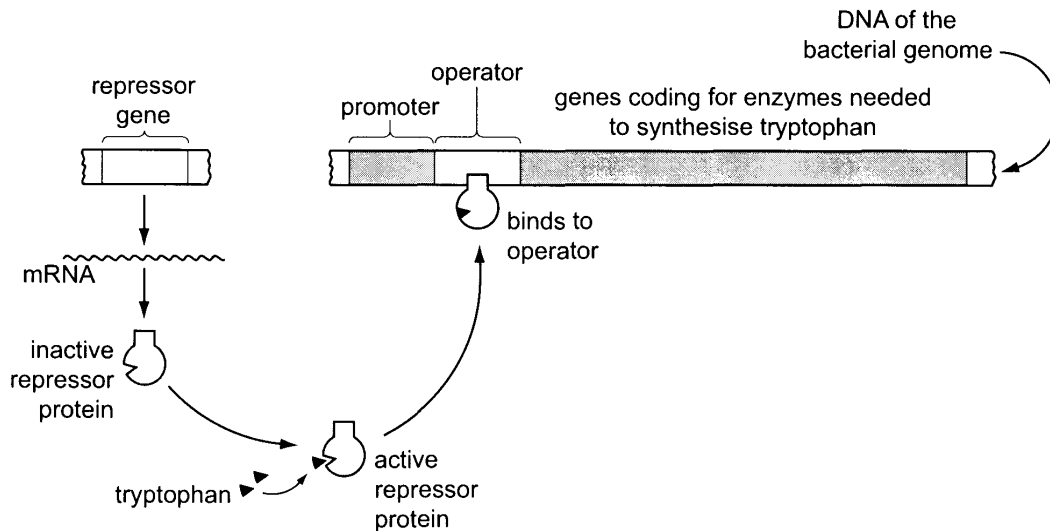
- A It blocks translocation of the ribosome along the mRNA.

B It interferes with chain termination and release of peptide.

C It prevents formation of the translation initiation complex.

D It inhibits binding of amino-acyl-tRNAs to the A site of the ribosome.

14 The diagram outlines the **control of tryptophan production** in *Escherichia coli*.



Which **aspect** of the control system in the diagram is **different from** the **lac operon** in *E. coli*?

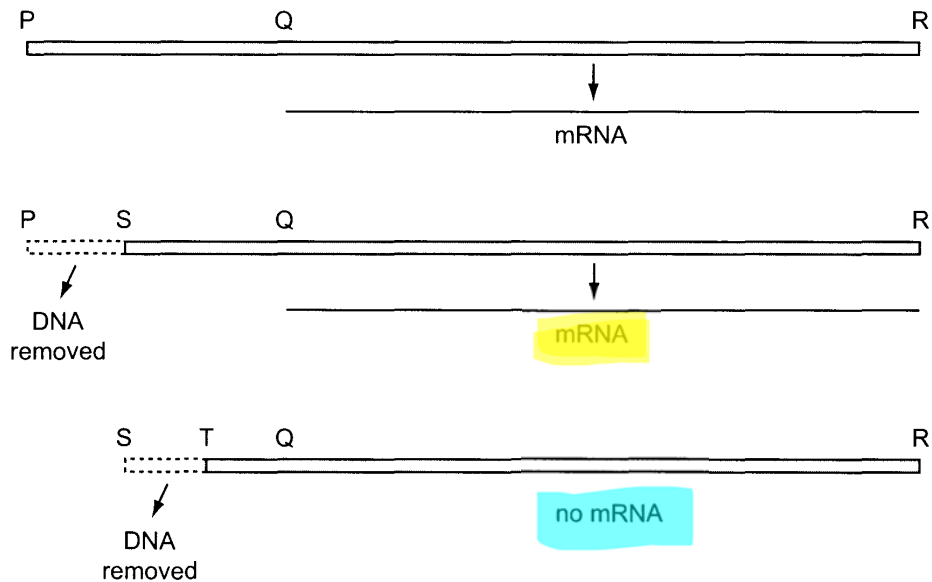
A operator blocked by active repressor protein

B presence of promoter

C repressor gene produces mRNA

D repressor protein activated by substrate

- 15 A length of DNA contains a non-coding region, PQ and a coding region, QR. The effect on the coding region of removing parts of the non-coding region was investigated. Results of removing firstly the length of DNA, PS and then ST are shown in the



Which **conclusion** may be deduced from these results?

- A Removal of any DNA from the non-coding region affects transcription.
- B The beginning of the promoter region is between P and S.
- C The beginning of the promoter region is between S and T.
- D The non-coding region has no role in transcription.

- 16 **Chromosomal mutations** were induced to produce a fertile hybrid species from cabbage and radish.

The table below shows the chromosome numbers in the parental species and the hybrids.

type of cell	number of chromosomes per cell
parental cabbage	18
parental radish	18
parental gametes	9
F ₁ hybrids	18
F ₁ gametes	9
F ₂ hybrids	18
F ₂ gametes	18
F ₃ hybrids	36

At **which stage** did the **chromosomal mutation occur**?

- A during the formation of the F₁ gametes
- B during the formation of the F₂ gametes**
- C during the fusion of the parental gametes
- D during the fusion of the F₁ gametes

17 Which statement about cancer is **not correct**?

- A Both alleles for a tumour suppressor gene must undergo loss-of-function mutations in the development of cancer.
- B Mutations in proto-oncogenes may not necessarily result in excessive cell proliferation.
- C Angiogenesis is a fundamental step in the development of a malignant tumour.
- D The risk of cancer is higher when *p53* gene is placed under the control of a strong viral promoter.

18 Which of the following statements could explain why a **combination of different drugs** rather than a single drug is being **used** to **treat HIV patients**?

- 1 HIV has a **short generation span** (eg. 2 days).
- 2 In an AIDS patient, the HIV infection produces **many new viruses per day** (eg. 10^{10} new viruses per day or more).
- 3 HIV has an RNA genome which has a higher mutation rate than DNA as it is single stranded.
- 4 The RNA genome allows the HIV to mutate rapidly to **acquire the resistance** to the drug being used.
- 5 Insertion of the HIV genome into the host cells will result in the **host cells mutations** that will confer resistance to the drug.

- A 1 and 2 only
- B 3 and 4 only
- C 1, 2 and 3 only
- D 3, 4 and 5 only

19 Some events that take place during **generalized transduction** are listed below.

- 1 Bacterial host DNA is fragmented
- 2 Bacterial DNA instead of viral DNA may be packaged in a phage capsid
- 3 Recombination between donor DNA and recipient DNA
- 4 Phage infects a bacterial cell
- 5 Phage DNA and proteins are made
- 6 Release of progeny virus

Which **sequence** of events is **most accurate** in describing generalized transduction?

First → last

A 4, 1, 3, 5, 6, 2

B 4, 1, 5, 2, 6, 3

C 4, 3, 1, 5, 2, 6

D 4, 5, 1, 3, 2, 6

20 Which statements **correctly describe** the **transfer of DNA** from one bacterium to another?

- 1 Both chromosomal and non-chromosomal DNA are transferred to daughter cells via binary fission.
- 2 Bacteriophage transfers DNA from one bacterium to another during specialised transduction.
- 3 Single strand of F plasmid transferred from one bacterium to another during conjugation.
- 4 Bacterial cell takes up foreign DNA from culture medium during generalised transduction.

A 1 and 2

B 2 and 3

C 1, 2 and 3

D 1, 3 and 4

- 21 The H1N1 virus is a subtype of the influenza A virus. Which statement about the H1N1 virus is **true**?

A The virus contains the **enzyme reverse transcriptase**.

B The virus comprises of a nucleocapsid enclosed by a membrane.

C The virus is **not** encased by a lipoprotein envelope.

D The virus genome is single-stranded **DNA**.

- 22 The table shows the blood group phenotypes resulting from crosses betw genotypes.

genotype of 1 st parent	genotype of 2 nd parent	
	I ^A I ^A	I ^A I ^O
I ^A I ^O	A	A and O
I ^B I ^O	A and AB	A, B, O and AB

Two parents have a **son** who has **blood group A** and **phenylketonuria**.

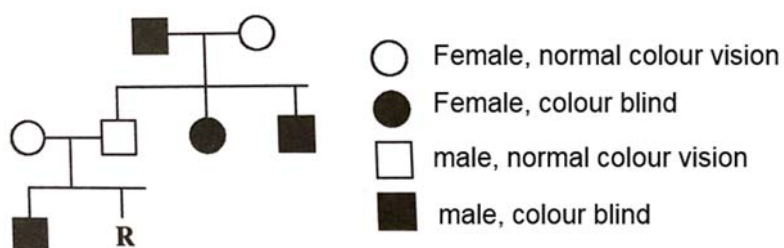
One **parent** has **blood group O** and the **other** has **blood group AB**.

Neither parent has **phenylketonuria**.

What is the **probability** that the **second child** of these parents will be a **girl** with **blood group B** who does **not have phenylketonuria**?

A 1 in 16 B 1 in 8 C 3 in 16 D 3 in 8

- 23 The diagram shows the inheritance of red-green colour blindness in a family. This condition is caused by a **sex-linked recessive allele**.



What is the **probability** that **individual R** will be a **colour-blind boy**?

A 0 B 0.125 C 0.25 D 0.5

- 24 A pure-breeding plant with the dominant phenotype of character **P** and the recessive phenotype of character **Q** was crossed with another pure-breeding plant with the recessive phenotype of character **P** and the dominant phenotype of **Q**.

The offspring of this cross were crossed with a double homozygous recessive for **P** and **Q** and the following results obtained:

22 were phenotypically dominant for **P** and recessive for **Q**.

5 were phenotypically dominant for both **P** and **Q**.

4 were phenotypically recessive for both **P** and **Q**.

24 were phenotypically recessive for **P** and dominant for **Q**.

Which one of the following types of inheritance is illustrated by these results?

- A gene linkage of **P** and **Q**
- B independent segregation of **P** and **Q**
- C Mendelian dihybrid inheritance
- D multiple alleles
- 25 In a comparative study of brinjal plants, a test cross was made between the variety of plant producing purple and long brinjal and the variety producing green and short brinjal. The results of the test cross are shown below:

Phenotypes	Number
Purple, Long	28
Purple, Short	30
Green, Long	26
Green, Short	34
Total number	118

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$\nu = c - 1$$

key to symbols

Σ = 'sum of ...'
 ν = degrees of freedom
 c = number of classes
 O = observed 'value'
 E = expected 'value'

	Expected numbers	Observed numbers	O-E
Purple, Long	29.5	28	
Purple, Short	29.5	30	
Green, Long	29.5	26	
Green, Short	29.5	34	

$$\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{(28-29.5)^2}{29.5} + \frac{(30-29.5)^2}{29.5} + \frac{(26-29.5)^2}{29.5} + \frac{(34-29.5)^2}{29.5}$$

$$= 0.076 + 0.008 + 0.415 + 0.686$$

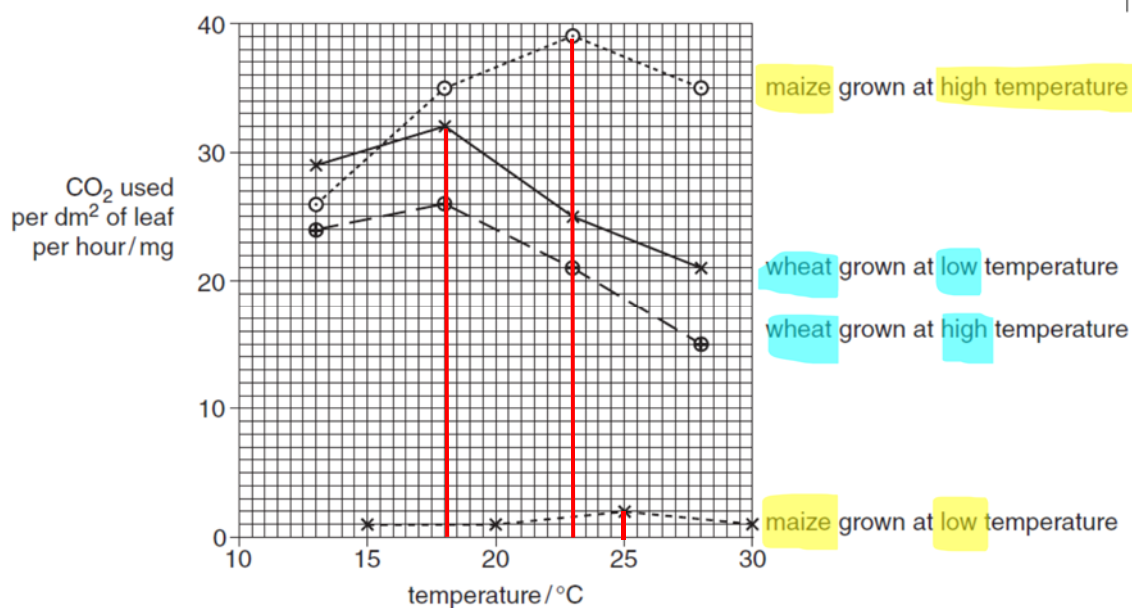
$$= 1.19$$

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Which of the following statements is **false**?

- A** The probability that the difference between observed and expected values is due to chance is greater than 10%.
- B** The two genes coding for the colour and shape are not linked.
- C** The difference between the expected number and the observed number of the phenotypes occurred by chance.
- D** The calculated χ^2 value is greater than the critical χ^2 value.

- 26** Young maize and wheat plants were grown to maturity at high and low temperatures. The rate of photosynthesis in each of these mature plants was measured at different temperatures. The rate of photosynthesis was measured as the amount of CO₂ used per dm³ of leaf per hour. The results are shown in the graph below.



What information can be **concluded** from the graph above?

- 1 For plants grown at high temperature, the rate of photosynthesis is optimum at 25°C in maize and 18°C in wheat.
- 2 For plants grown at high temperature, maize had a greater increase in rate of photosynthesis compared to wheat until optimum temperature was reached.
- 3 The rate of photosynthesis was affected more significantly in maize plants than in wheat plants grown at low temperatures.
- 4 Low temperatures slowed down the formation of membranes in maize plants but not in wheat plants which caused a decrease in lamellae formation.

A 2 and 3 only

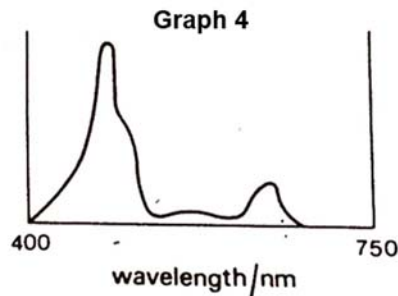
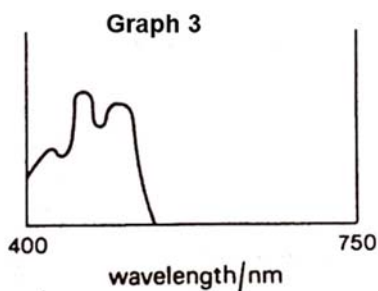
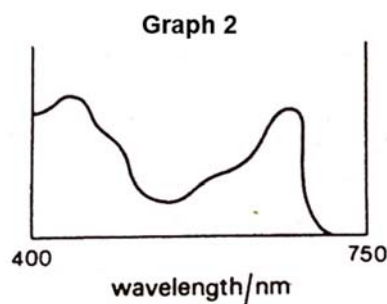
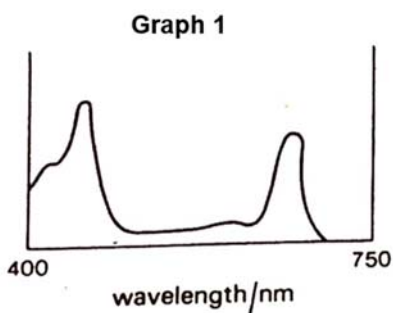
B 1 and 4 only

C 1, 2 and 3 only

D All of the above

- 27 Three of the four graphs below show the **absorption spectra** of photosynthetic pigments. The other graph shows the action spectrum of photosynthesis for a plant containing those pigments.

The x-axes show wavelength. Three of the y-axes show the percentage light absorbed while the other shows the rate of photosynthesis measured by the rate of evolution of oxygen.



Which one of the following series **identifies** the **four graphs**?

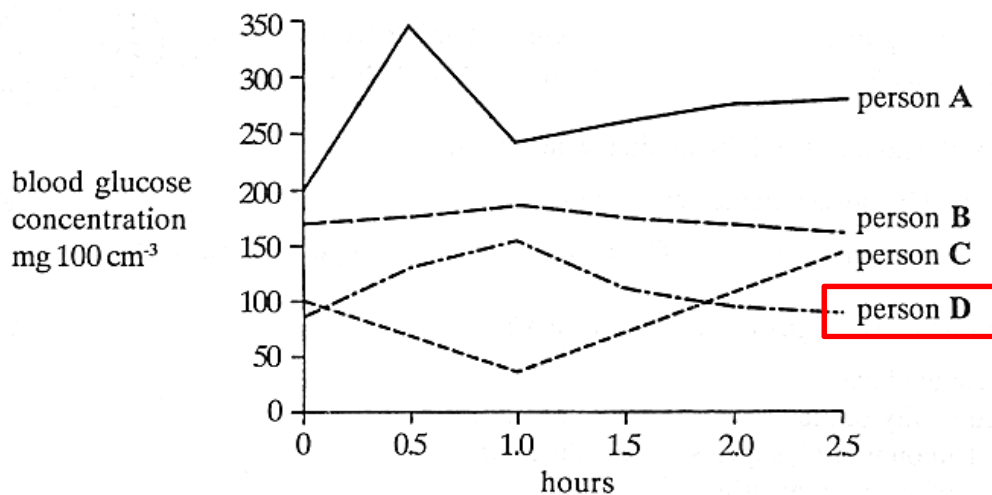
	<i>Absorption spectra</i>			<i>Action spectrum</i>
	Chlorophyll a	Chlorophyll b	Carotenoids	
A	1	4	3	2
B	2	1	3	4
C	4	2	1	3
D	3	2	4	1

28 Which hormones **promote** the **processes** shown?

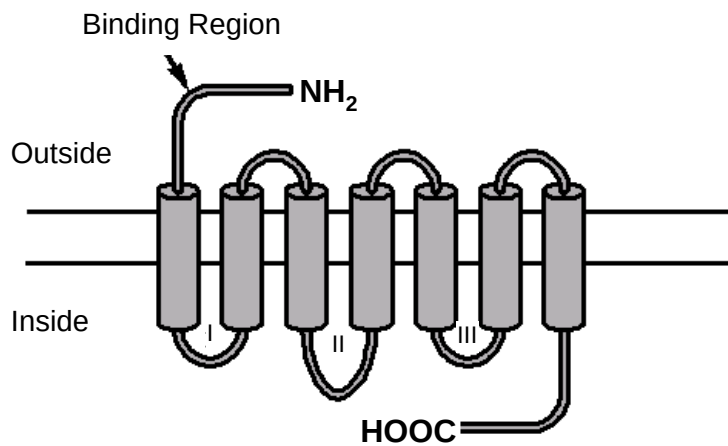
	conversion of glycogen to glucose in liver cells	respiration of glucose in liver cells	uptake of glucose by muscle cells
A	insulin	glucagon	glucagon
B	insulin	glucagon	insulin
C	glucagon	insulin	glucagon
D	glucagon	insulin	insulin

29 The blood glucose concentration of four people was **measured** over a **2.5 hour period** **after drinking** a **concentrated glucose solution**. The graph shows the results.

Which person is **least likely** to have **problems** associated with **insulin secretion**?



30 The figure below shows a protein receptor on a cell membrane.



Which statement **correctly describes** how this **receptor** responds **upon binding** to a **ligand**?

- A It enters the nucleus, undergoes a conformational change and binds to DNA.
- B It undergoes a conformational change, enters the nucleus and binds to DNA.
- C It undergoes conformational change and binds to G-protein.**
- D It undergoes dimerisation, and hence a conformational change.

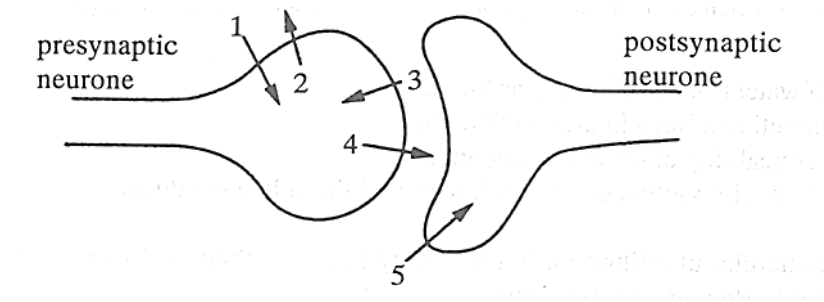
- 31 Insulin is released from cells in the islets of Langerhans while acetylcholine is released from neurons.

Both events take place in response to changes in cytosolic concentration of calcium ions.

Which of the following correctly describes the release of insulin and acetylcholine?

	Release of insulin	Release of acetylcholine
A	Arrival of an action potential causes Ca^{2+} to enter the cytosol of β -cells No action potential involved in stimulating β -cells	Signal transduction causes Ca^{2+} to enter the cytosol of neurons Action potential cause Ca^{2+} to enter neurons
B	Signal transduction causes Ca^{2+} to enter the cytosol of β -cells	Arrival of an action potential causes Ca^{2+} to enter the cytosol of neurons
C	Arrival of an action potential causes Ca^{2+} to exit the cytosol of α -cells No action potential involved It should be β -cells	Signal transduction causes Ca^{2+} to exit the cytosol of neurons Action potential involved in stimulating neurons Ca^{2+} to enter not exit
D	Signal transduction causes Ca^{2+} to exit the cytosol of α -cells	Arrival of an action potential causes Ca^{2+} to exit the cytosol of neurons

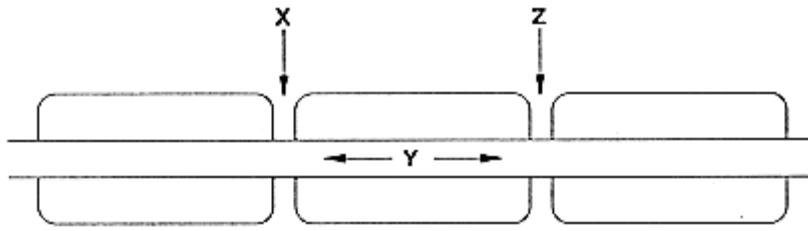
- 32 The diagram shows the sequence of events occurring as an **action potential arrives** at a **synapse**. The numbered arrows represent the movement of substances across the membrane.



What are the **substances moving across** the **membrane**?

	1	2	3	4	5
A	K ⁺	Na ⁺	acetylcholine	Ca ²⁺	K ⁺
B	K ⁺	Na ⁺	K ⁺	Ca ²⁺	acetylcholine
C	Na ⁺	K ⁺	Ca ²⁺	acetylcholine	Na ⁺
D	Na ⁺	K ⁺	Na ⁺	acetylcholine	Ca ²⁺

- 33 The diagram shows a myelinated axon.



How does the **myelin sheath increase the speed of impulse transmission** along the axon?

- A It ensures that the ions are kept close to the axon membrane in region Y.
 - B It insulates the axon, **so** increasing the potential at regions X and Z.
 - C It restricts the change in potential difference to regions X and Z.**
 - D It promotes a change in potential difference in region Y.
- 34 **Northern elephant seals, *Mirounga angustirostris*, were nearly hunted to extinction** in the 1890s, with only about **20 individuals left** at the end of the century.

The **population** has now **grown** to **more than 120 000**.

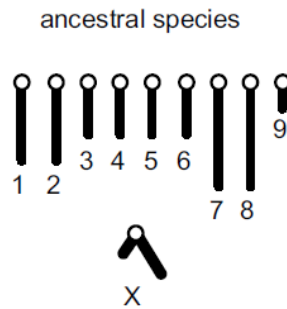
In the 1890s, southern elephant seals, *Mirounga leonina*, were not as severely hunted and currently there are estimated to be 600 000 southern elephant seals.

Based on this information, it is **true** to say that

- A northern elephant seals have evolved as a result of the 'founder effect'.
- B northern elephant seals would show less genetic variation than southern elephant seals.**
- C southern elephant seals would have experienced greater genetic drift than northern elephant seals.
- D the mutation rate in northern elephant seals would have been greater than in southern elephant seals.

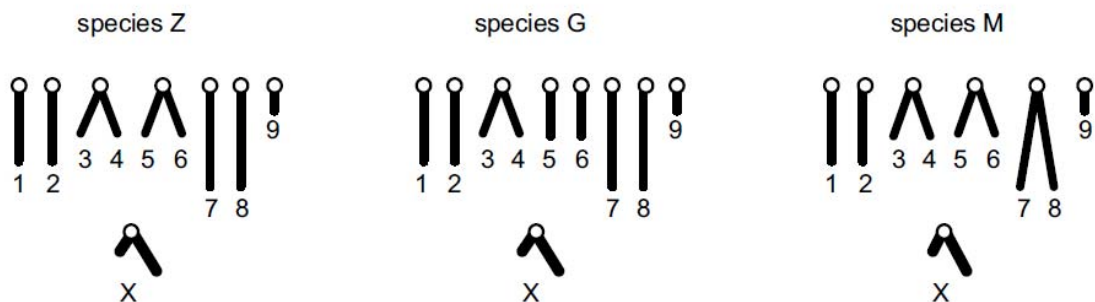
- 35** In some Australian insects, new species have arisen through changes that occurred to chromosomes in an ancestral species. Such changes may involve the joining together of chromosomes, the loss of whole or parts of chromosomes, and rearrangement of the genetic material within chromosomes.

One **ancestral species** has the following **haploid set of chromosomes**.



As the changes in chromosomes accumulate, a number of different species can result from a single ancestral species.

Three species that have **evolved from** the **ancestral species** shown above have the haploid sets of chromosomes shown below.



The **most likely order of evolution** of these species is

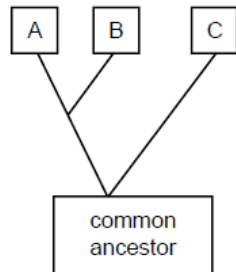
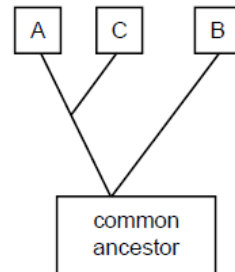
- A** ancestral species, species Z, species G, species M.
- B** ancestral species, species G, species M, species Z.
- C** ancestral species, species M, species G, species Z.
- D** ancestral species, species G, species Z, species M.

- 36** Cytochrome-c is a protein found in most organisms. The amino acid sequence of this protein varies between species. The **number of differences** in the amino acid sequences in the protein of cytochrome-c between three species of chordate A, B and C are shown in the table below.

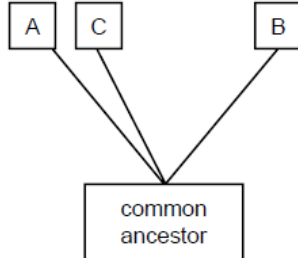
	Species B	Species C
Species A	11	3
Species B		10

Based on this evidence, the **phylogenetic tree** that **best represents** the possible **evolutionary relationships** between the **three species** is

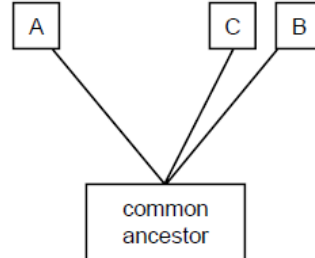
A

**B**

C



D



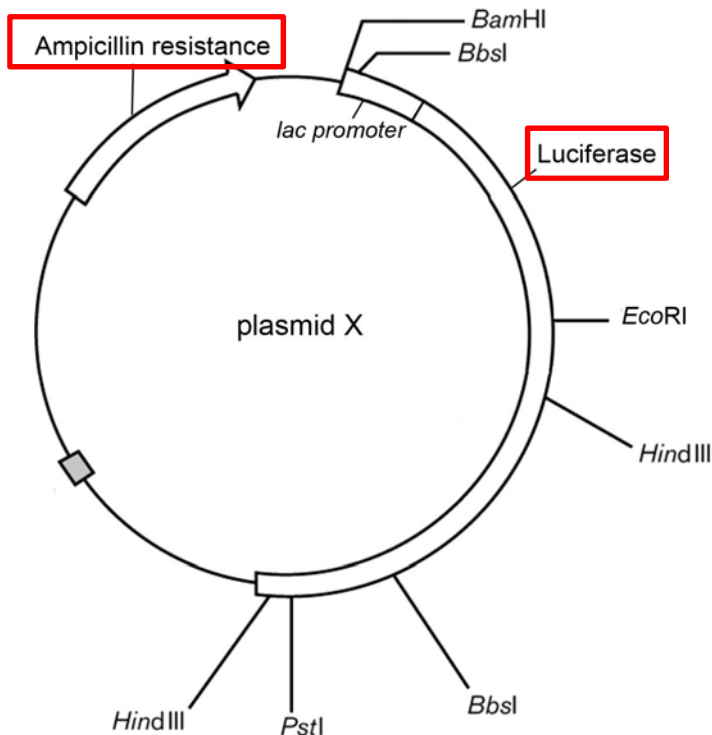
- 37 Corn is a major crop grown in Europe. In the past, it was either ruined by attack from the corn borer larvae or intensively sprayed with pesticides each year. The biotech company Novartis gained approval to insert a *Bt* gene into corn. This gene codes for a protein that kills the larvae feeding on the corn. The genetically engineered *Bt* corn initially thrived without the addition of any pesticides, but later suffered damage from pests again.

Which of the following are **possible reasons** why the *Bt* corn is **again** under **attack from pests**

- 1 A strain of resistant larvae has emerged as a chance mutation. The frequency of the gene conferring resistance as well as the **number of larvae with resistance** has **decreased**.
- 2 Corn is being attacked by other pests which are not killed by the protein encoded by the *Bt* gene.
- 3 Mutations in corn may have led to the loss of the *Bt* allele.

- A 1 and 2 only
 B 1 and 3 only
C 2 and 3 only
 D All of the above

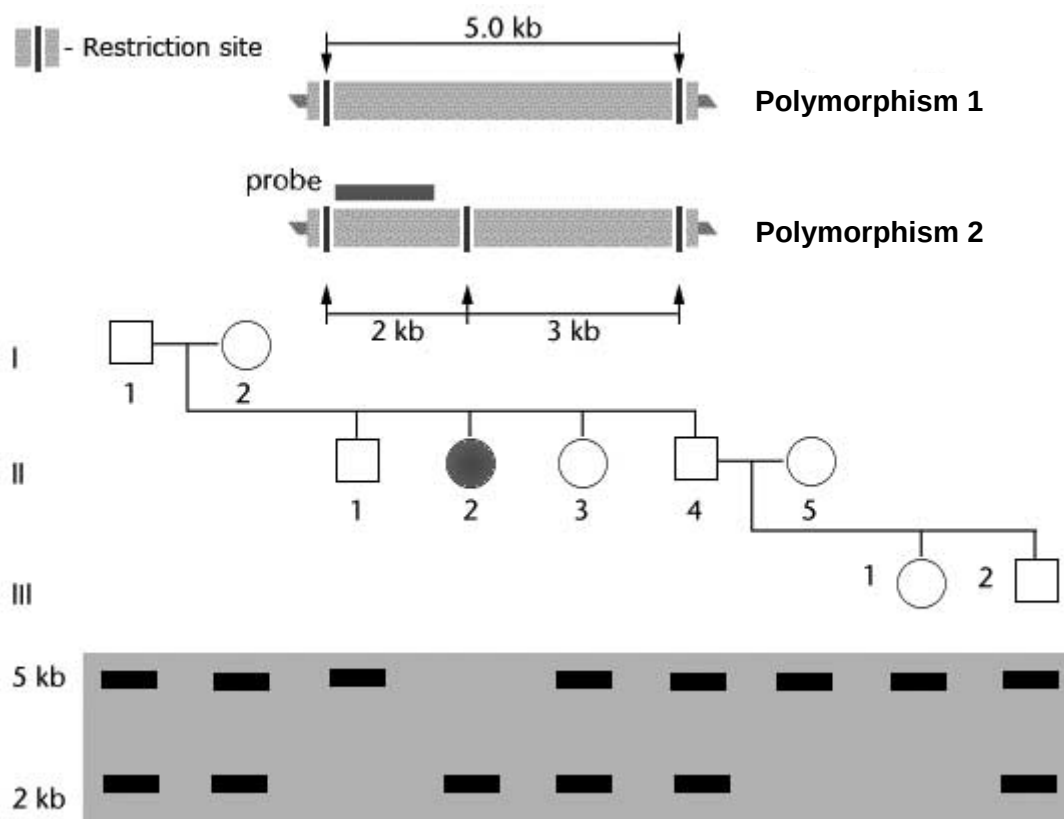
- 38 Plasmid X can serve as a **vector** for the insertion of genes to be cloned.



Which of the following options will allow the selection of the colonies containing the recombinant form of plasmid X?

	Selection medium	Phenotype of colonies that contain the inserted gene
A	Containing ampicillin and lactose	White colonies
B	Containing ampicillin and luciferase	Colonies that emit light
C	Containing ampicillin, lactose and luciferin	White colonies
D	Containing ampicillin, lactose and luciferin	Colonies that emit light

- 39 Restriction Fragment Length Polymorphism (RFLP) was used to analyse the inheritance of a rare mutation that causes the pre-mature degeneration of the nervous system. A genetic marker on Chromosome 5 was discovered to be associated with the disease and used in the analysis. Genomic DNA was digested with a restriction enzyme *Bam*HI and the results of the gel electrophoresis and nucleic acid hybridisation are shown in the diagram below.



Which of the following observations **cannot be made** from the information provided?

- A The mutation that causes the disease is probably recessive.
- B The mutation that causes the disease is likely to be located in a coding region of the genome.
- C The children of II-1 are unlikely to inherit the disease.
- D There can only be a maximum of three *Bam*HI restriction sites on Chromosome 5 in the affected individual.**

- 40 Non-viral *ex vivo* transfer of a gene encoding coagulation factor VIII was performed. Stem cells isolated from the bone marrow of patients suffering from severe hemophilia A were transfected with the gene encoding coagulation factor VIII.

What is the **sequence of events** for the ***ex vivo* transfer of the gene** encoding coagulation factor VIII?

- 1 Transfection with vectors containing the gene encoding coagulation factor VIII.
 - 2 Implantation of cells into patients.**
 - 3 Isolation of stem cells.
 - 4 Selection and cloning of cells expressing coagulation factor VIII.
- A 2, 3, 1, 4
 - B 3, 1, 4, 2**
 - C 3, 2, 4, 1
 - D 3, 4, 2, 1